Introduction to Programming and Software Engineering

NGUYEN Duc Dung, Ph.D.
Institute of Information Technology, VAST

Informatics: Course Plan

- History of Computing and Computer Architecture *(no practice)*
- Operating System
- Software Tools for Techies
- Data Presentation and File Structures
- Networks and the Internet
- **Introduction to Programming and Software Engineering**

Programming Languages

Hello.c

```c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
```

Hello.py

```python
# Hello World program in Python

print ("Hello World!");
```


Programming Languages

Hello.c

```c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
```

Hello.pas

```pascal
Program HelloWorld(output);

begin
  writeln('Hello, world!');
end.
```


Hello.c

```c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
```

Hello.py

```python
# Hello World program in Python

print ("Hello World!");
```

Hello.pas

```pascal
Program HelloWorld(output);

begin
  writeln('Hello, world!');
end.
```


Main Functions

Hello.c

```c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
```

Hello.pas

```pascal
Program HelloWorld(output);

begin
  writeln('Hello, world!');
end.
```

Variables: 65 + 190 = ?

```c
#include <stdio.h>

int main()
{
    char c = 'A';
    int i = 0;

    i = i + 1;
    printf("i = %d\n", i);
    printf("c = %c; c = %d\n", c, c);
    // 65 + 190 = ?
    c = c + 190;
    printf("c = %d\n", c);
    return 0;
}

```

7

Variables: 1 + 0.01 =?

C Program

```c
#include <stdio.h>

int main()
{
    int i = 1;
    i = i + 0.01;
    printf("i = %d\n", i);

    return 0;
}
```

Python Program

```python
i = 1
i = i + 0.01
print(i)
```

8

Variables

C Program

```c
#include <stdio.h>

int main()
{
    int i = 1;
    i = i + 0.01;
    printf("i = %d\n", i);

    i = "USTH";
    printf("i = %d\n", i);
    printf("i = %s\n", i);

    i = i + 1;
    printf("i = %s\n", i);

    return 0;
}
```

Python Program

```python
i = 1
i = i + 0.01
print("i = ", i)

i = 'USTH'
print("i = ", i)
#printf("i = %s\n", i);

i = i + 1
print("i = ", i)
```

9

Variables: 1 + 0.01 =?

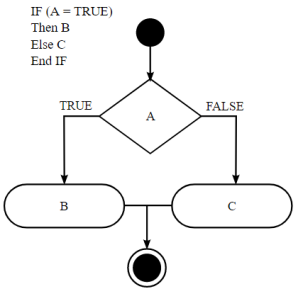
```python
# Hello World program in Python
# print "Hello World!\n"

i = 0
i = i + 1
i = i + 0.01

print(i)
```

10

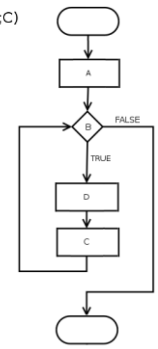
IF …THEN…ELSE



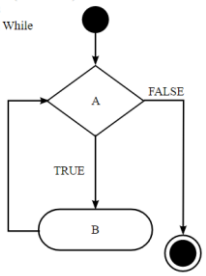
11

FOR, WHILE LOOP

for(A;B;C)
D;



While (A = TRUE) Do
B
End While



12

Process Data Automatically

No.	Studet ID	Family Name	Given Name	Speciality	Attendant	Midterm	Final exam	Final mark
1	BA10-006	Lê Mai	Chi	PMAB	9	8	7	
2	BA10-020	Nguyễn Ngọc	Hoa	PMAB	7	9	8	
3	BA10-024	Kiều Khánh	Huyền	CS	8	7	6	
4	BA10-025	Nguyễn Thị Khánh	Huyền	ICT	9	4	9	
5	BA10-026	Nguyễn Thị Khánh	Lê	FST	8	5	5	
6	BA10-042	Đỗ Thảo	Nhi	PMAB	8	2	5	
7	BI11-051	Vũ Hoàng Minh	Đức	PMAB	7	5	6	
8	BI11-070	Phan Nhật	Duy	AE	9	7	7	
9	BI11-075	Bùi Thị Thu	Hằng	WEO	6	8	8	
10	BI11-081	Phan Gia	Hiền	PMAB	5	9	7	

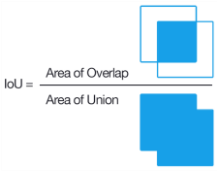
13

Homework

- **Solving the quadratic equation**

  $$Ax^2 + Bx + C = 0$$

  - Input: A, B, C
  - Output: Solution of the equation
- Calculating the **intersection over the union** of two rectangles
  - Input: (x1, y1, w1, h1), (x2, y2, w2, h2)
  - Output: IoU value



14